

# Assignment1: CancerMortality

Joan Acero, Anna Monsó, Walter Troiani

2024-10-21

## Contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>Introduction</b>                                             | <b>2</b>  |
| <b>First Steps</b>                                              | <b>3</b>  |
| <b>Explanatory Data Analysis</b>                                | <b>5</b>  |
| Univariate Descriptive Analysis & Data Quality Report . . . . . | 5         |
| Data Quality Report . . . . .                                   | 15        |
| Imputation . . . . .                                            | 17        |
| Profiling & Relationships . . . . .                             | 19        |
| <b>Modeling</b>                                                 | <b>20</b> |
| Initial Model exploration and Multicollinearity study . . . . . | 20        |
| Model transformations . . . . .                                 | 21        |
| Influential data study . . . . .                                | 22        |
| Further model exploration . . . . .                             | 22        |
| Model validation . . . . .                                      | 22        |
| Model evaluation . . . . .                                      | 22        |

# Introduction

**This report does not include all the code and outputs produced throughout the project but highlights the key ones to support the explanations provided. For a complete view of all outcomes that informed the conclusions presented in this report, please refer to the accompanying RMarkdown file.**

This report documents the process and results of the Modeling Assignment for the Statistical Modeling and Inference course. The primary aim of this project is to analyze the Cancer Mortality dataset and build an accurate and interpretable linear model to predict `target_deathrate`, which represents the average cancer mortality rate per 100,000 individuals.

The project begins with data preparation and exploratory analysis, focusing on cleaning and enhancing the dataset to ensure its quality. This involves generating a Data Quality Report, identifying and handling missing values through imputation, and analyzing variable distributions and relationships. Univariate descriptive analysis is performed to examine each variable's characteristics.

The second stage involves modeling using the training dataset, where several key analyses are conducted to optimize the model:

- Initial model exploration addresses potential issues such as multicollinearity and the contribution of each predictor.
- Model transformations are applied to improve variable relationships and address non-linearity.
- An influential data study identifies and evaluates the impact of unusual observations on the model.
- The model is further refined through validation and diagnostics, ensuring it adheres to key assumptions like normality, homoscedasticity, and independence.

Finally, the evaluation process assesses the model's generalization performance on the test dataset, providing insights into its predictive capabilities. A detailed interpretation of the model results is presented, highlighting how the predictors influence the target variable and their practical implications for understanding cancer mortality trends.

This report follows a systematic and methodical approach to ensure the development of a reliable model.

## First Steps

To begin, we import the necessary data, load the required libraries, and create functions to streamline repetitive processes.

```
calculate_outliers <- function(data) {  
  # Check for regular outliers  
  Boxplot(variable, main = "Boxplot")  
  outliers <- boxplot.stats(data)$out  
  
  varsumm <- summary(data)  
  Q1 <- varsumm[2]  
  Q3 <- varsumm[5]  
  IQR <- Q3 - Q1  
  
  lower_severe_bound <- Q1 - 3 * IQR  
  upper_severe_bound <- Q3 + 3 * IQR  
  lower_mild_bound <- Q1 - 1.5 * IQR  
  upper_mild_bound <- Q3 + 1.5 * IQR  
  
  # Identify outliers  
  mild_outliers <- which((data < lower_mild_bound) | (data > upper_mild_bound))  
  severe_outliers <- which((data < lower_severe_bound) | (data > upper_severe_bound))  
  num_mild_outliers <- length(mild_outliers)  
  num_severe_outliers <- length(severe_outliers)  
  
  # Return the results as a list  
  return(list(  
    num_outliers = num_mild_outliers,  
    num_severe_outliers = num_severe_outliers,  
    outliers_indexes = mild_outliers  
  ))  
}  
  
summary_and_normality_check <- function(data) {  
  #Basic summary  
  print("Summary: ")  
  print(summary(data))  
  
  # Plot normality  
  print("Mean")  
  mm <- mean(data, na.rm = TRUE); print(mm)  
  print("Standard deviation")  
  ss <- sd(data, na.rm = TRUE); print(ss)  
  hist(data, freq = F, 100, main = "Histogram")  
  curve(dnorm(x, mean = mm, sd = ss), col="red", lwd = 2, add=T)  
  
  #Compute skewness  
  print("Skewness")  
  print(skewness(data, na.rm = TRUE))  
  
  #Number of outliers  
  out.res <- calculate_outliers(data)
```

```

print("Num Outliers:")
print(out.res$num_outliers)
print("Num. severe outliers:")
print(out.res$num_severe_outliers)

#Shapiro test
print("Shapiro test: ")
print(shapiro.test(data))

# Most repeated value:
freq_table <- table(variable)
most_repeated_value <- names(freq_table)[which.max(freq_table)]; most_repeated_value
most_repeated_count <- max(freq_table); most_repeated_count
print("Most repeated value ad times it occurs: ")
print(most_repeated_value)
print(most_repeated_count)

return(out.res$outliers_indexes)
}

outliers_df <- df
outliers_df[,] <- 0

```

# Explanatory Data Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the dataset, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test dataset.

## Univariate Descriptive Analysis & Data Quality Report

In this initial phase, we analyze all variables in the dataset, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones.

Subsequently, we present a data quality report that assesses the overall quality of the data and discusses how it may affect the modeling process.

### Variable *avganncount*

*avganncount* is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” First, note that it contains 126 observations with the value “1962.667684”, which we consider inconsistent and that represent NAs. After setting this individuals as NAs, we start the variable analysis.

The difference between the mean (524.24) and the median (155) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (74 and 383, respectively) are much lower than the mean and the maximum value of 38150, further suggesting skewness. As expected, the Shapiro-Wilk normality test gets rejected. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 207 mild outliers and 131 severe outliers.

```
variable <- df$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
variable[variable == 1962.667684 ] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

### Variable *avgdeathsperyear*

*avgdeathsperyear* is a numerical variable representing the “Average number of reported cancer-related mortalities per year.”

This variable is complete, with no missing values. Descriptive statistics reveal a mean of 191.6 and a median of 62, indicating a strongly non-symmetric distribution. This asymmetry is further supported by the first quartile (29) and third quartile (140.5), both significantly lower than the mean, suggesting a right-skewed distribution. Visual exploration using a boxplot confirms this skewness, displaying a pronounced right tail.

Additionally, a histogram with an overlaid normal curve underscores the substantial deviation from normality, a conclusion reinforced by the results of the Shapiro-Wilk test. The distribution exhibits a significant number of outliers, with 225 classified as mild and 138 as severe, all of which are above the third quartile. These

outliers contribute to the variable's extended right tail and may warrant further investigation to understand their context and potential implications.

```
variable <- df$avgdeathsperyear
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "avgdeathsperyear"] <- 1
```

### Variable *target\_deathrate* :

*target\_deathrate* is the numeric target variable of the dataset. While a more detailed analysis will be presented in subsequent sections, we introduce key insights regarding its distribution, missing values, and outliers here.

The target variable is complete in the training sample, with no missing values. The median (178.3) and mean (178.8) are nearly identical, suggesting a symmetric distribution. This symmetry is further supported by visual inspections of the boxplot and histogram, both of which confirm the balanced nature of the data around its center.

To evaluate normality, we performed the Shapiro-Wilk test. The resulting small p-value rejected the null hypothesis, indicating that the distribution deviates from normality despite its apparent symmetry. Thus, the target variable is not normally distributed.

In terms of outliers, the variable contains 35 observations classified as outliers, including one severe case.

```
variable <- df$target_deathrate;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "target_deathrate"] <- 1
```

### Variable *incidencerate* :

*incidencerate* is a numerical variable representing the “Mean per capita cancer diagnosis rate per 100,000 individuals.”

Anomalies were identified in the data: the same 126 individuals who had a value of “1962.667684” in the *avganncount* variable also showed a value of “453.5494221” in the *incidencerate* variable. These values are considered non-coherent, and thus, we set them as missing (NA). After setting these values to NA, we proceed with the analysis of the column's distribution.

The mean of the variable is 448.7, the median is 448.7, and the standard deviation is 57.85, indicating moderate variability in the data. The skewness of 1.40 suggests a positively skewed distribution, with the tail extending toward higher values. This is further confirmed by the presence of 36 outliers, including 4 severe outliers, which are all above the upper threshold. The Shapiro-Wilk strongly rejects the null hypothesis of normality.

These results suggest that while the variable is mostly symmetric within its central range, the skewness and presence of outliers may have implications for modeling and analysis.

```
variable <- df$incidencerate;
l11 <- which(variable == 453.5494221 ); length(l11)
variable[variable == 453.5494221 ] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "incidencerate"] <- 1
```

### Variable *medincome* :

*medincome* is a numerical variable representing the “Median income per county.” The distribution is neither normal nor symmetric, as evidenced by the histogram, Shapiro-Wilk test, and skewness value (1.47), all indicating a right-skewed distribution with a long tail. Descriptive statistics show a minimum of 22,640, a maximum of 122,641 (Loudoun County, Virginia), and a mean of 47,278 with a standard deviation of 12,342. The median income is 45,454, with the first and third quartiles at 39,031 and 52,612, respectively. There are no missing values, but the variable contains 69 outliers, 15 of which are severe, all above the upper quartile. These findings suggest substantial variability and a pronounced concentration of counties with lower incomes compared to a smaller group with exceptionally high incomes.

```
variable <- df$medincome;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "medincome"] <- 1
```

### Variable *popest2015* :

*popest2015* is a numerical variable representing the population of each county in 2015.

It contains no missing values but has an extensive range and an extremely skewed distribution, as highlighted by a skewness value of 14.71 and a Shapiro-Wilk test confirming strong deviation from normality. The distribution is heavily right-skewed, with a mean of 106,841 significantly exceeding the median of 27,158, and a wide spread between the 1st quartile (12,191) and 3rd quartile (66,880). The variable includes 252 mild outliers and 162 severe outliers, all exceeding the upper whisker of the boxplot. The highest value, 10,170,292, corresponds to Los Angeles County, California. The standard deviation of 376,494 reflects the substantial variability, driven by a small number of extremely populous counties. This skewed distribution highlights the vast disparity in county populations, with most counties having smaller populations and a few extreme outliers significantly influencing the overall statistics.

```
variable <- df$popest2015;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "popest2015"] <- 1
```

### Variable *povertypercent* :

*povertypercent* is a numerical variable representing the percentage of the population in poverty. It has no missing values and a moderate positive skew, as indicated by the skewness value of 0.89 and the Shapiro-Wilk test, confirming the variable does not follow a normal distribution. The mean is 16.79, slightly higher than the median of 15.7, reflecting the slight skew. The 1st quartile (12.15) and 3rd quartile (20.4) show a moderate spread, with values ranging from a minimum of 3.7 to a maximum of 44. The standard deviation is 6.41, indicating variability in poverty levels across counties.

The variable has 42 mild outliers, but no severe ones. The most frequent value is 14.7, occurring 22 times. While this variable shares a non-normal distribution with other numerical variables like *medincome* and *incidence*, its skewness and outlier characteristics are less pronounced, indicating a more balanced distribution overall.

```
variable <- df$povertypercent;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "povertypercent"] <- 1
```

### Variable *studypercap* :

*studypercap* is a numerical variable representing the “Per capita number of cancer-related clinical trials per county.”

A significant feature of this variable is that 1,162 out of 1,831 counties (63.5%) have a value of 0, making this the most frequent value and dominating the distribution. Nearly all non-zero values are flagged as outliers due to their extreme deviation from the majority, with 307 identified as mild outliers, and 222 severe ones.

Given the predominance of zero values and the highly skewed nature, this variable may contribute more noise than predictive value to the model. **Removing it from both the training and test datasets is a reasonable decision to simplify the model and enhance its performance.**

```
variable <- df$studypercap;
res.var <- summary_and_normality_check(variable)

outliers_df[res.var, "studypercap"] <- 1
```

### Variable *binnedi* : I CONSIDER IT IS WORTH TO REMOVE

Variable *binnedi* is a categorical variable (or factor) that expresses the median income per capita binned by decile.

```
variable <- df$binnedinc;
summary <- summary(variable); summary
table(variable)
prop.table(table(variable))
barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```

### Variable *medianage* :

Medianage is a numerical variable that expresses the median age of county residents.

The variable has no missing values. Has 65 outliers, 18 of which are severe outliers. All of these outliers should be removed because it's impossible to be more than 100 years. Once we removed all the incorrect values we get 52 outliers, 18 of them severe. Finally we observe a symmetric distribution with a median = 40.90 and a mean=40.85. After analyzing the boxplot, and the histogram along the Shapiro test we can discard the normality of the variable's distribution.

```
variable <- df$medianage
table(variable)
variable[variable > 70 ] <- NA
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "medianage"] <- 1
```

### Variable *medianagemale* :

Numeric variable, represents the mean age of the male population of each county.

There are no missing values and it has 46 outliers, non of them several outliers. The mean is 39.59 yo and the median 39.50yo what suggest a simmetry distribution. After looking at the results of the shapiro-test it doesn't have a normal distribution

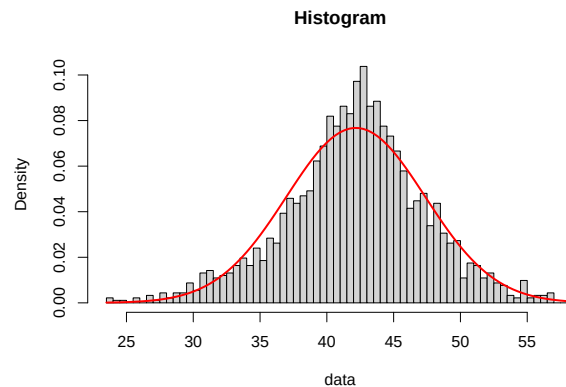
```
variable <- df$medianagemale
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"medianagemale"] <- 1
```

**Variable *medianagemale* : I AM NOT SURE IF IT'S WORTH IT TO SEPARATE THE MEDIANAGE WITH SEX**

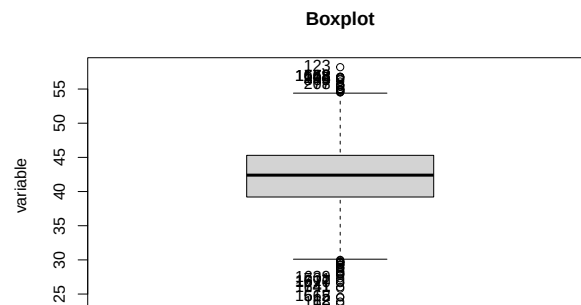
The medianagemale is a numerical variable that analyze the median age of the females in each county. It has a simmetric distribution (median=42.40, mean=42,17) but after analyzing the results of the Shapiro-test we observe that does not have a normal distribution. There are 55 outliers none of them severe

```
variable <- df$medianagemale
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  23.60  39.20   42.40   42.17  45.30   58.20
## [1] "Mean"
## [1] 42.16931
## [1] "Standard deviation"
## [1] 5.196007
```



```
## [1] "Skewness"
## [1] -0.2057034
```



```
## [1] "Num Outliers:"
## [1] 55
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99321, p-value = 1.817e-07
##
## [1] "Most repeated value ad times it occurs: "
## [1] "42.9"
## [1] 31
```

```
outliers_df[res.var,"medianagefemale"] <- 1
```

### Variable *geography* :

Categorical variable that shows the name of each county. We can use this variable as a PK

–comment – Maybe we can separate the name of the state and create a new variable to be able to study the distribution around the United States

```
variable <- df$geography;
summary <- summary(variable); summary
table(variable)
```

### Variable *percentmarried*

Numerical variable that represent the percent of county residents who are married.

It has a non-symmetric distribution, it is skewed with a long left tail observed in the histogram. It doesn't have missing values, it has 34 outliers (non of them severe) and they are concentrated on the lower end of the distribution. With the shapiro-test we can confirm the rejection of the null hypothesis, therefore it doesn't have a normal distribution.

```
variable <- df$percentmarried
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"percentmarried"] <- 1
```

### Variable *pctnohs18\_\_24*

Numerical variable that show the percentage of county residents aged 18-24 whose highest level of education is less than a high school diploma.

The distribution of this variable is skewed with a long right tail, we can confirm that it doesn't have a normal distribution thanks to the histogram and the curve. Moreover the p-value obtained in the shapiro-test confirm the non normal distribution. It doesn't have missing values, and it has 35 outliers, 5 of them severe.

```
variable <- df$pctnohs18_24
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctnohs18_24"] <- 1
```

### Variable *ppcths18\_24*

Numeric variable, it represent the percentage of county residents ages 18-24 highest education attained is a high school diploma.

We can observe with the median(34,7%) and the mean(35%) that the variable has a symmetric distribution, but after the shapiro\_test we observe that it doesn't have a normal distribution. It doesn't have missing values and it has 33 outliers, non of them severe.

```
variable <- df$pcths18_24
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pcths18_24"] <- 1
```

### Variable *pctsomecol18\_24*

Numerical variable. Percent of county residents ages 18-24 highest education attained some college.

It has 1375 missing values. (75%) therefore we are going to do further actions in the imputation part. We will obtain the values knowing that  $pctnohs18\_24 + pcths18\_24 + pctsomecol18\_24 + pctbachdeg18\_24 = 100$

– A: Hauriem de fer els calculs aquí i ja calcular els outliers i tal be o ho fem a la part de imputation com esta ara?

```
variable <- df$pctsomecol18_24
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctsomecol18_24"] <- 1
outliers <- which(outliers_df$pctsomecol18_24 == 1); length(outliers)
```

### Variable *pctbachdeg18\_24*

Numerical variable. Percent of county residents ages 18-24 highest education attained is a bachelor's degree. There are no missing values, it has 56 mild outliers 10 of them severes. After the shapiro-test we can reject the null hypothesis, thus it doesn't have a normal distribution.

```
variable <- df$pctbachdeg18_24
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctbachdeg18_24"] <- 1
```

### Variable *pcths25\_over*

Numerical variable that express the percentage of county residents ages 25 and over highest education attained a high school diploma.

It doesn't have missing values but it have 18 outliers, non of the severe. It doesn't have a normal distribution.

```
variable <- df$pcths25_over
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pcths25_over"] <- 1
```

### Variable *pctbachdeg25\_over*

Numerical variable that expresses the the percentage of country resident ages 25 and over highest education attained: Bachelor's degree.

Firstly we can see that there is a huge left skewness and there is a relatively large number of outliers (62), from which just 3 are severe. From the plot we can already tell that no normal distribution is followed and a shapiro test over the variable confirms it.

– op2 : mild = 59 | sev = 3

```
res.var <- summary_and_normality_check(df$pctbachdeg25_over)
outliers_df[res.var, "pctbachdeg25_over"] <- 1
```

### Variable *pctemployed16\_over*

Numerical variable that represents the percentage of county residents ages 16 and over, employed.

This variable contains over 82 NA values and has a slight right skewness and 11 mild outliers. Both inferential and visualization methods clearly point that the real distribution is extremely unlikely to be normal.

```
res.var <- summary_and_normality_check(df$pctemployed16_over)
outliers_df[res.var, "pctemployed16_over"] <- 1
```

### Variable *pctunemployed16\_over*

Numerical variable that represents the percentage of county residents ages 16 and over, unemployed.

This variable is heavily left skewed and contains over 46 outliers, of which 4 are severe. Despite the distribution looking almost normal there is a big number of outliers and a long heavy right tail. Applying a shapiro test we reject the hypothesis that the distribution is normal.

– op2: mild = 42 | sev = 4

```
res.var <- summary_and_normality_check(df$pctunemployed16_over)
outliers_df[res.var, "pctunemployed16_over"] <- 1
```

### Variable *pctprivatecoverage*

Numerical variable that represents the percentage of county residents with private health coverage.

Despite having no NA and just 17 mild outliers, this variable has a huge standard deviation and a right skewed mean. Applying a shapiro test we reject the hypothesis that the distribution is normal.

```
res.var <- summary_and_normality_check(df$pctprivatecoverage)
outliers_df[res.var, "pctprivatecoverage"] <- 1
```

**Variable *pctprivatecoveragealone*** We should check if the imputation is good enough to decide if we remove the variable or we keep it

Numerical variable that represents the percentage of county residents with private health coverage alone (no public assistance at all).

The number of NA is humongous (356), but it has a small amount of mild outliers (4) and almost no skewness at all. Both using inferential and visualizations methods we cannot easily reject the hypothesis that this variable is not normal. So we can loosely say that is **normally** distributed.

– op2 : After doing the Shapiro-Wilk test and obtaining a  $p\_value = 0,14$  we can not reject the null hypothesis, therefore this variable has a normal distribution.

```
res.var <- summary_and_normality_check(df$pctprivatecoveragealone)
outliers_df[res.var, "pctprivatecoveragealone"] <- 1
```

**Variable *pctempprivcoverage***

Numerical variable that represents the percentage of county residents with employee-provided private health coverage.

Despite the low number of outliers (7 mild), the absence of NA, the small left skewness and the median being almost equal to the mean, we should reject the null hypothesis that the distribution is normal with a 5% significance. However, with a smaller significance level than 3% we couldn't reject that this distribution is not normal.

– we should clarify which is the C.level at the begining of the project and then justify this part

```
res.var <- summary_and_normality_check(df$pctempprivcoverage)
outliers_df[res.var, "pctempprivcoverage"] <- 1
```

**Variable *pctpubliccoveragealone***

Numerical variable that represents the percent of county residents with government-provided health coverage alone.

This variable contains 21 mild outliers, a great left skewness and a long right tail. Performing an inferential test like shapiro we get a tiny p-value which makes us reject the hypothesis that the distribution is normal under any significance level.

```
res.var <- summary_and_normality_check(df$pctpubliccoveragealone)
outliers_df[res.var, "pctpubliccoveragealone"] <- 1
```

**Variable *pctwhite***

Numerical variable that represents the percent of county residents who identify as White.

Both from the plot and from inferential methods we can easily discard that this variable is anywhere near normal, being a hugely right skewed variable. It contains 104 outliers from which 7 are severe.

This points out that the vast majority of counties have a high percentage of white population.

– op2 : mild = 97 | sev = 7

```
res.var <- summary_and_normality_check(df$pctwhite)
outliers_df[res.var, "pctwhite"] <- 1
```

### Variable *pctblack*

Numerical variable that represents the percent of county residents who identify as Black.

There is a huge number of outliers (323), from which 99 are severe and there is also a huge left skewness, that points out that the vast majority of counties (except the outliers) have a rather low percentage of black folks. The distribution almost looks like a power law distribution (Zipf) therefore we can safely reject both with visualization and inferential methods that the distribution of this variable is not normal.

– op2: mild = 224 | sev = 99

```
res.var <- summary_and_normality_check(df$pctblack)
outliers_df[res.var, "pctblack"] <- 1
```

### Variable *pctasian*

Numerical variable that represents the percent of county residents who identify as Asian.

Pretty similarly to the last distribution, we have a heavily left skewed zipfian like distribution, with 305 outliers (from which 107 are severe), pointing out that the vast majority of counties have less than 3% of asian percentage.

– op2: mild = 198 | sev = 107 – A: no acabo de entendre el resultat obtingut al boxplot dsp de mirar el histogarma

```
res.var <- summary_and_normality_check(df$pctasian)
hist(df$pctasian , 50 )
outliers_df[res.var, "pctasian"] <- 1
```

### Variable *pctotherrace*

Numerical variable that represents the percent of county residents who identify as any other race (Pacific islanders, arábians, saiayjins, namekians...).

Again this variable, just like the last 2 presents a huge left skewness and zipfian like distribution with 283 outliers, from which 102 are severe. Again this sampled distribution is not normal.

– op2 : mild = 181 | sev = 102

```
res.var <- summary_and_normality_check(df$pctotherrace)
outliers_df[res.var, "pctotherrace"] <- 1
```

### Variable *pctmarriedhouseholds*

Numerical variable that represents the percent of married households.

This variable contains 59 outliers from which 2 are severe and a noticeable left skewness. Visualization isn't that clear, but using inferential methods like shapiro test we reject the null hypothesis that this variable is normal.

op2: mild = 57 | sev = 2

```
res.var <- summary_and_normality_check(df$pctmarriedhouseholds)
outliers_df[res.var, "pctmarriedhouseholds"] <- 1
```

### Variable *birthrate*

Numerical variable that represents the number of births relative to number of women in that specific county.

Finally, this last variable contains a fair amount of outliers (121) from which 17 are severe. It also has a fair amount of left skewness. Both from visualization and shapiro testing we reject that this sample variable is normal.

op2 : mild = 104 | sev = 17

```
res.var <- summary_and_normality_check(df$birthrate)
outliers_df[res.var, "birthrate"] <- 1
```

## Data Quality Report

Now that all the variables have been slightly analyzed in terms of their meaning. Given that univariate outliers and NA values have already been computed and shown, only a ranking and a visual summary will be showcased to gather the main insights visually.

This quality report will be divided between variables and individuals of the dataset, and finally a multivariate outlier analysis will be performed.

### Per variable

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers.
- Rank variables according the sum of missing values (and errors).

```
# Number of errors (including inconsistencies). TODO
md.pattern(df, rotate.names=T)
```

On one hand, looking at the pattern, we can notice that there are 362 variables with no null values, and that the vast majority of nulls (1056) are located on the pctsomecol18\_24 variable with just 1 null column. Another remarkable trend is that up to 248 individuals have 2 null values in variables pctsomecol18\_24 and pctprivatecoveragealone, while this other variable just has 83 pure nulls (Only this column). The last variable that contains a low amount of null values is pctemployed16\_over, where 20 individuals have the last 3 mentioned variables containing nulls, and the rest of few possible NA values come from pattern permutations of these 3 variables. Therefore this dataset only has 3 possible rows containing NA, and only one of them is relevant in this aspect. Ranking:

| Rank | Null values                     |
|------|---------------------------------|
| 1    | pctsomecol18_24: 1375 NA        |
| 2    | pctprivatecoveragealone: 356 NA |
| 4    | avganncount : 126 NA            |

| Rank | Null values               |
|------|---------------------------|
| 4    | incidencerate : 126 NA    |
| 3    | pctemployed16_over: 82 NA |

On the other hand, the top 5 ranking for variables with the most amount of mild and severe outliers is:

*TO\_DO -> CHECK*

| Rank | Mild Outliers    | Severe Outliers      |
|------|------------------|----------------------|
| 1    | avganncount 333  | avganncount 257      |
| 2    | pctblack 323     | studypercap 222      |
| 3    | studypercap 307  | popest2015 162       |
| 4    | pctasian 305     | avgdeathsperyear 138 |
| 5    | pctotherrace 283 | pctasian 107         |

## Per individuals

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

There are 362 obs without NA in any variable, 1144 obs with 1 variable with a NA value, 305 obs with 2 NA values and 20 obs with 3 NA values along the 33 variables of the dataset.

```
#Number of missing values
na_counts <- rowSums(is.na(df))
table(na_counts)
```

Out of all observations, 971 do not exhibit any outliers across the variables, while 385 observations contain a single outlier in one variable. Interestingly, there is 1 observation where we identified outliers in 10 variables and another 10 observations that are outliers in 8 variables.

```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
```

Considering over 20 numerical variables that contain few correlated variables and considering a 1% significance level, 209 multivariate outliers are found in the dataset.

```
#Identify individuals considered as multivariant outliers.
df$avgdeathsperyear <- as.numeric(df$avgdeathsperyear)
df$medincome <- as.numeric(df$medincome)
df$popest2015 <- as.numeric(df$popest2015)

#Compute multivariant outliers and find their ids (24)
subset_num_vars <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
res_mout <- Moutlier(df[,subset_num_vars], quantile=0.97, plot=T)
```

```

mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))

#Reset graphical parameters and plot the outliers
dev.off()
plot(x=res_mout$md, y=res_mout$rd, main="Classic Mahalanobis D vs Robust Mahalanobis D (M. Outliers)",
abline(v=res_mout$cutoff, col="red")
abline(h=res_mout$cutoff, col="red")

```

We can clearly see that for the number of outliers variable, the mean is around 1.39, and the third quantile around 2.0, however, we have a high standard deviation of 1.935 pointing out at the presence of outliers in the number of outliers variable, suggesting there are several individuals with more than 2 outlier variables. This variable contains somewhat high correlation with pcths25\_over, pchwhite, pctasian and other weak correlations.

The number of missing values doesn't make much sense to be analyzed after removing 2 of the main problematic columns which contained several null values.

Finally, looking at the multivariate factor profiling, we see that there is a global association between "mout" and "binnedinc", and we can see that there is a huge global association with quantitative variables pctotherrace and n\_out (Logically, the number of univariate outliers contributes to a possible multivariate outlier individual), followed by medianage looking at their eta squared values. Finally, we see looking at the mean values of the quantitative variables that in the case of an individual that is not a multivariate outlier, almost every single variable mean value is equal or close to the overall mean. However in the case of multivariate outliers we see huge discrepancies in the mean of pctasian, pctotherrace, n\_out, popest2015, avgdeathspereyear and avganncount, which are way higher than the overall mean value. This may point out that the asian-otherrace populations and those with higher population or more average deaths per year, are more prone to be multivariate outliers.

```

anal_df <- df
#Binary factor if its a multivariate outlier
anal_df$mout <- 0
anal_df[mout_ids, "mout"] <- 1
anal_df$mout <- factor(anal_df$mout, labels=c("Not M.O", "Yes M.O"))
#Number of univariate outlier variables for a given observation
anal_df$n_out <- rowSums(outliers_df)
#Number of missing values in variables for a given observation
anal_df$n_miss <- rowSums(is.na(df))

summary(anal_df[, c("mout", "n_out", "n_miss")]); sd(anal_df$n_out); sd(anal_df$n_miss)
num_vars <- anal_df[sapply(anal_df, is.numeric)]
corrplot(cor(num_vars))

catdes(anal_df, num.var=34, proba=0.05)

```

## Imputation

### pctsomecol18\_24

Given the huge amount of null values, the impute PCA has not been able to replicate the original distribution of this messy variable.

```

num_vars <- df[sapply(df, is.numeric)]
miss_ids <- which(is.na(df$pctsomecol18_24))

```

```

res_imp <- imputePCA(num_vars)
imp_pctsomecol <- res_imp$completeObs[, "pctsomecol18_24"]

# Calculate the quantiles for validation
quantiles_df <- quantile(df$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctsomecol, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol"),
      col = c("blue", "red"), lwd = 2)

#We won't impute the values!
df[miss_ids, "pctsomecol18_24"] <- imp_pctsomecol[miss_ids]

# --- op2 ---
summary(df$pctsomecol18_24)
imp_values <- 100 - df$pctnohs18_24[miss_ids] - df$pcths18_24[miss_ids] - df$pctbachdeg18_24[miss_ids]
pct2 <- df$pctsomecol18_24
pct2[miss_ids] <- imp_values

quantiles_2 <- quantile(pct2, probs = seq(0,1,0.1), na.rm = FALSE)
summary(pct2)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")

lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2);
lines(seq(0, 1, 0.1), quantiles_2, col = "green", lwd = 2);
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol", "imputation2"),
      col = c("blue", "red", "green"), lwd = 2)

# add imputed values
df$pctsomecol18_24 <- pct2

```

## pctemployed16\_over

pctemployed16\_over only contains 70 null values and the imputePCA function has been able of almost perfectly preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

```

num_vars <- df[sapply(df, is.numeric)]
miss_ids <- which(is.na(df$pctemployed16_over))
res_imp <- imputePCA(num_vars)
imp_pctemployed16_over <- res_imp$completeObs[, "pctemployed16_over"]

# Calculate the quantiles for validation
quantiles_df <- quantile(df$pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = FALSE)

```



povertypercent, pctpubliccoverage, pcths25\_over and pctunemployed16\_over. Looking at the inverse, we see a medium-high inverse correlation with variables medincome, pctbachdeg25\_over and pctprivatecoverage and pctmarriedhouseholds.

```
subset_num <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
subset_cat <- c(9)
subset_df <- c(subset_num, subset_cat)

res_anal <- condes(df[, subset_df], num.var=3, proba=0.01)
res_anal
```

## Serial correlations

It has been found previously that the target\_variable doesn't follow a normal distribution at all, now its clear looking at the auto correlation function plot below that this variable contains serial correlation. To address this issue we will

```
library(stats)
acf(df$target_deathrate)

ids <- sample(1:nrow(df), nrow(df))
df <- df[ids, ]
acf(df$target_deathrate)
```

# Modeling

## Initial Model exploration and Multicollinearity study

Firstly, it is important to realize that just as in past sections, the geography variable should just be used as a supplementary variable and not used for modelling purposes. Two initial models are built, one with all the numeric variables, and a second one containing all the numeric variables + all categorical variables.

Both models explain roughly 51% and 52% of variance respectively (R2), and both have multiple significant slope coefficients such as percentmarried, pctbachdeg25\_over, pctemployed16\_over, pctotherrace, avganncount, incidencerate, avgdeathspereyear and binnedinc level 7 among the ones with smallest p-values.

Looking at variance over inflation of the first model, we can see that there are several variables that contain a great amount of variance over inflation, pointing out the presence of multicollinearity. The main ranked variables that exceed the high VIF threshold (10) in both models:

1. avgdeathspereyear
2. pctpubliccoverage
3. popest2015
4. pctprivatecoveragealone
5. pctprivatecoverage
6. pctpubliccooveragealone
7. avganncount
8. percentmarried
9. binnedinc

Exploring the residual analysis plots of each model, a common trend is easily identified. Due to the presence of extreme outliers, both models do not meet the linear model premises at all:

1. Residuals vs Fitted plots look like residuals are random and do not follow any pattern except for a couple of extreme outliers, which distort the straight line, therefore we cannot claim linearity.
2. Q-Q Residuals suggest that the residuals almost follow a normal distribution (are random), despite a few outlier and heavier tails than usual, therefore we cannot fully claim normality.
3. Scale-Location plots show a slight curvature due to the outlier values that points out a possible small heteroscedasticity.
4. Residuals vs Leverage confirms that despite almost all points have a quite low leverage, there are a small amount of highly leveraged points that distort the model in general (Specially observation 600).

```
m1 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p
summary(m1)
vif(m1)
par(mfrow=c(2,2)) # Change the panel layout to 2 x 2
plot(m1, id.n=0)

# Mixed (Categorical + numerical)
m2 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p
summary(m2)
vif(m2)
plot(m2)
```

Following up, we will build m3, a modification of m2 removing some inflating variables (avgdeathsperyear, pctprivatecoveragealone, pctpubliccoveragealone, binnedinc, percentmarried, medianagefemale, medianagemale) which some of them are redundant. This is one approach for addressing multicollinearity in the model, another strategy could be to perform a dimensionality reduction technique like PCA. The results we get from this variable removal is a model with all the independent variables having a less than 10 VIF (However, we still have variables that are above 5 VIF like avganncount, popest2015...), yet still retaining over 50 R-squared. However, the residual analysis results are pretty much the same as in the past 2 models with minor improvements. Outliers keep distorting the model heavily. Anova nested model test confirms that is worth using the lighter model.

```
m3 <- lm(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypercent + stu
vif(m3)
summary(m3)
plot(m3)

#anova(m3, m2)
```

One further improvement for a following model would be to remove variables with high VIF and high p-value in the coefficient tests, like popest2015 or avganncount. Another possible improvement would be to retain binnedinc instead of its redundant counterpart medincome.

```
library(effects)
plot(allEffects(m3))
```

## Model transformations

Possibly we need to perform some transformations...

```
#Exemple random ;)  
#boxcox(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypercent + stud  
#boxTidwell(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypercent + .
```

## Influential data study

Given that the models are being quite negatively affected by influential data, an study of the aforementioned will be performed to address this issue.

```
#Include Code
```

## Further model exploration

```
#Include Code
```

## Model validation

In this section we will assess how well the model performs on the training data (train split) to ensure its robustness and capabilities, so we can assess under/over fitting.

We will analyze the residuals to ensure that model assumptions are met and use R-squared as the main performance metric will the model will be computed by means of Ordinary least squares.

```
#Include Code
```

## Model evaluation

Finally we will evaluate how well the model performs with unseen data (train split) to assess its forecasting capabilities, using the same R-squared performance metric.

```
#Include Code
```